■■ Sentinel Security Agent: Simple Step-by-Step Guide

A Plain Language Explanation of How Sentinel Works & Which Method Does What

1. What is Sentinel?

Sentinel is your personal AI security guard for your Gmail inbox. It automatically scans incoming emails and immediately tells you if an email is **SAFE (Low Risk)**, **SUSPICIOUS (Medium Risk)**, or a **DANGEROUS PHISHING SCAM (High Risk)**.

2. How Sentinel Works (Step-by-Step)

Step 1: You Open the Sentinel App

You open the web dashboard in your browser. It displays your overall security health score and threat counters.

Step 2: You Click 'Connect your Gmail'

Sentinel connects safely to Google using OAuth 2.0. It only asks for **read-only permission** to check email text. It never sees your password or sends emails.

Step 3: Sentinel Fetches Your Latest Emails

Every few seconds, Sentinel asks Gmail for your latest 20 emails. It uses parallel processing (Promise.all) so your emails load in under 2 seconds.

Step 4: Sentinel Runs 5 AI Security Checks

The backend sends the email text to Sentinel's Python AI brain. The AI evaluates sender legitimacy, suspicious links, urgency keywords, machine learning probabilities, and fuzzy logic scores.

Step 5: You See Clear Risk Results on Screen

Emails are color-coded in real-time:

- ■ **LOW RISK (Green):** Safe emails (Google Alerts, Bank Statements, Newsletters)
- ■ **MEDIUM RISK (Amber):** Slightly suspicious links or urgency
- ■ **HIGH RISK (Red):** Dangerous phishing, fake document lures, password traps

3. Which Method Works Where? (File & Code Guide)

File / Location	Method Used	What it Does in Simple Terms
frontend/src/App.jsx	React Dashboard	Shows the user interface, risk colors, live threat table, and buttons.
backend/server.js	Node.js Gateway API	Connects to Gmail, fetches 20 emails fast, and forwards text to Python AI.
ai/ai_server.py	Master Decision Server	Combines ML, Rules, Bayes, Fuzzy Logic, and Whitelists into 1 final risk score.
ai/security_rules.py	Forward-Chaining Rules	Looks for keyword traps like 'sign in', 'access document', and suspicious links.
ai/train_model.py	Machine Learning Trainer	Teaches the AI using TF-IDF word patterns and Logistic Regression.
ai/predict.py	ML Classifier	Gives an ML phishing probability score (0% to 100%) for new emails.

ai/bayes_risk.py	Naive Bayes Math	Calculates word statistics (scam frequency vs safe email frequency).
ai/fuzzy_risk.py	Fuzzy Logic Engine	Evaluates gray-area risks like urgency tone and domain trust.

4. How Sentinel Avoids Mistakes

- **No False Positives:** Genuine service emails (Google Security Alerts, IPPB Bank, ZeroSSL) are protected by a *Trusted Service Whitelist*, so real security notices are never flagged as phishing.
- **No False Negatives:** Clever phishing lures (like 'Sign in to access document') are caught by the *Rule Engine* even if the text looks clean, ensuring password traps are always flagged **HIGH RISK**.